

Uniform(1,2) Canonical Simulation Update

Purpose

This simulation restarts the fixed-DGP IFEval-like comparison using weaker nonzero loadings and an apples-to-apples canonical parameterization. The earlier Uniform(2,3) loading design was useful as a high-signal case, but it can make binary responses nearly deterministic for low and high latent classes. The new run uses milder loadings to diagnose whether factor-score collapse and parameter discrepancies are driven by saturation of the binary observations.

Simulation Design

For each design cell, the structural parameters are fixed and a new binary data set is simulated in each replication. The loading matrix uses the IFEval-like unbalanced block allocation, with at least 30 primary items in the smallest block. Primary and cross-loading magnitudes are both sampled from Uniform(1, 2), cross-loadings occur with probability 0.05, and cross-loading signs are randomly assigned.

Dimension	Values
Sample size	$n \in \{100, 200, 400\}$
Items, Product MAP	$p \in \{500, 1000, 1500, 2000\}$
Items, Gibbs baselines	$p \in \{500, 1000\}$
Latent factors	$H \in \{5, 10\}$
Mixture components	$G_h = 2$ for all h , or $G_h = 3$ for all h
Mixture separation	$\text{sep} = 2$
Replications	25 per setting

Product MAP and Viroli Laplace Gibbs use the same loading shrinkage level in matched cells:

$$\lambda_{\Lambda}(n) = \begin{cases} 3, & n \in \{100, 200\}, \\ 6, & n = 400. \end{cases}$$

The diffuse Gaussian Viroli baseline has no Laplace loading penalty.

Canonicalization

All reported parameter RMSEs use the same canonical factor scale. For each factor coordinate, the fitted marginal mixture is transformed to have mean zero and variance one. If

$$\tilde{f}_{ih} = \frac{f_{ih} - m_h}{s_h},$$

then loadings and intercepts are transformed so that the probit linear predictor is unchanged:

$$\alpha_j + \sum_{h=1}^H f_{jh} \lambda_{jh} = \tilde{\alpha}_j + \sum_{h=1}^H \tilde{f}_{jh} \tilde{\lambda}_{jh}.$$

Product MAP is canonicalized after fitting before evaluation. Viroli Gibbs normalizes each retained posterior draw using the same convention before posterior averaging. This makes factor, loading, intercept, mixture-mean, and mixture-variance RMSEs comparable across methods.

Computation Order

The run is staged deliberately.

1. Product MAP is run first for the full p -grid. Replications are launched serially, while each fit uses 18 internal workers for conditionally independent updates.
2. After Product MAP completes, the Gibbs baselines are run for $p \in \{500, 1000\}$. Independent Gibbs jobs are launched in parallel, and each Gibbs fit uses 4 internal workers for conditionally independent blocks. The Markov chain iterations themselves remain serial.

This separates the fast deterministic estimator from the more expensive Gibbs baselines and avoids oversubscribing the machine while Product MAP is using the full worker pool.

Recorded Outputs

The combined results file records timing, convergence flags, factor RMSE, probability-score RMSE, intercept RMSE, loading RMSE, mixture mean RMSE, mixture variance RMSE, mixture weight RMSE, and Gibbs effective sample-size summaries when available.

The run can be reproduced from the repository root with:

```
zsh scripts/sample_size/run_uniform12_canonical_product_first_simulation.sh
```

The active output directory is: `results/full/fixed_ifeval_lambda_min30_u1_2_cp0_05_sep2_npenalty3_6_h5_h10_canonical_productfirst_gibbsparallel/`

The final combined CSV will be: `results/full/fixed_ifeval_lambda_min30_u1_2_cp0_05_sep2_npenalty3_6_h5_h10_canonical_productfirst_gibbsparallel/comparison_results.csv`